

Welcome to the Data Science course!

Today is a Course introduction.



MODULE 1: COURSE OBJECTIVES.

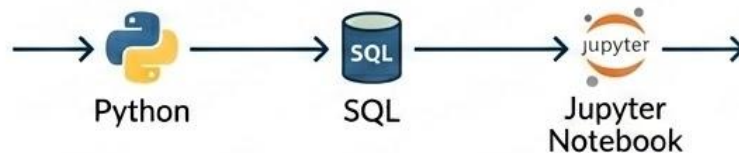
Define what is Data Science and why it is important

1. DEFINE DATA SCIENCE



Understand its core function and business importance.

2. UNPACK THE TOOLS



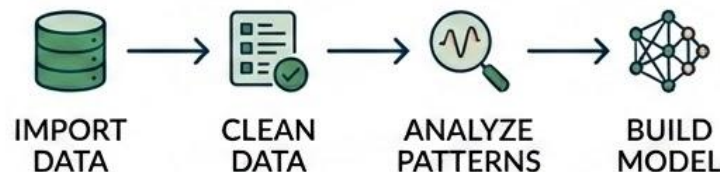
Analyze the critical tools used in modern data workflows.

3. BUILD YOUR ENVIRONMENT



Seamlessly set up your personal GitHub Codespace.

4. MAP THE JOURNEY

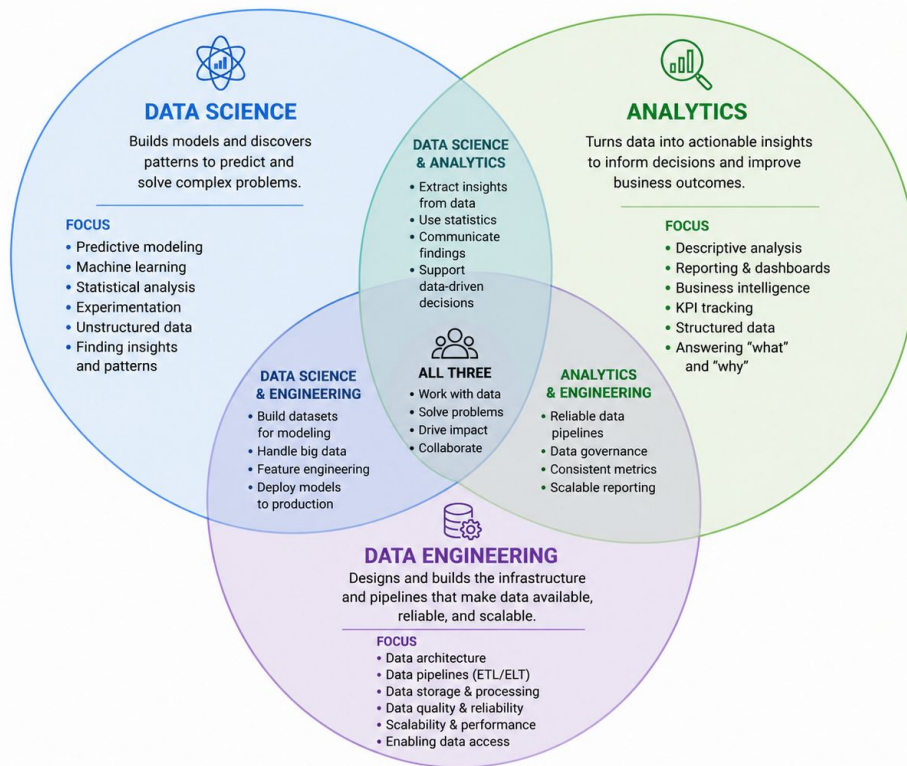


Preview the skills and topics of the full course.

What is data science?

DATA SCIENCE, ANALYTICS, AND ENGINEERING

Different Focus. Shared Goals. Complementary Skills.



What is Data Science?

At its core, Data Science is about using data to **understand the world**, **answer questions**, and **solve problems**.

It combines skills from mathematics, statistics, and computer programming to analyze information, find patterns, and make predictions.



Streaming Recommendations

How does your favorite service know what to recommend?



Online Shopping

Why do stores suggest products that you might like?



Scientific Research

How do scientists make sense of millions of pieces of data?

All of these are powered by some level of Data Science.

What Data Scientists Do

Data Scientists use tools to clean messy datasets, visualize information in meaningful ways, and create models to make decisions or predictions.

In this course, we'll take a hands-on approach to explore the field of Data Science using **Python**.



Business Optimization

Deciding the best location for a new store using spatial data analysis.



Healthcare Insights

Analyzing the effects of a new medicine to improve patient outcomes.



Computer Vision

Improving how self-driving cars recognize and react to obstacles.

Tackle real-world problems with modern Data Science tools.

A Journey for Everyone



Who is it for?

Data Science is for **anyone** curious about the world and eager to use data to make sense of it.

Whether you're interested in business, science, art, or technology, Data Science has something to offer.



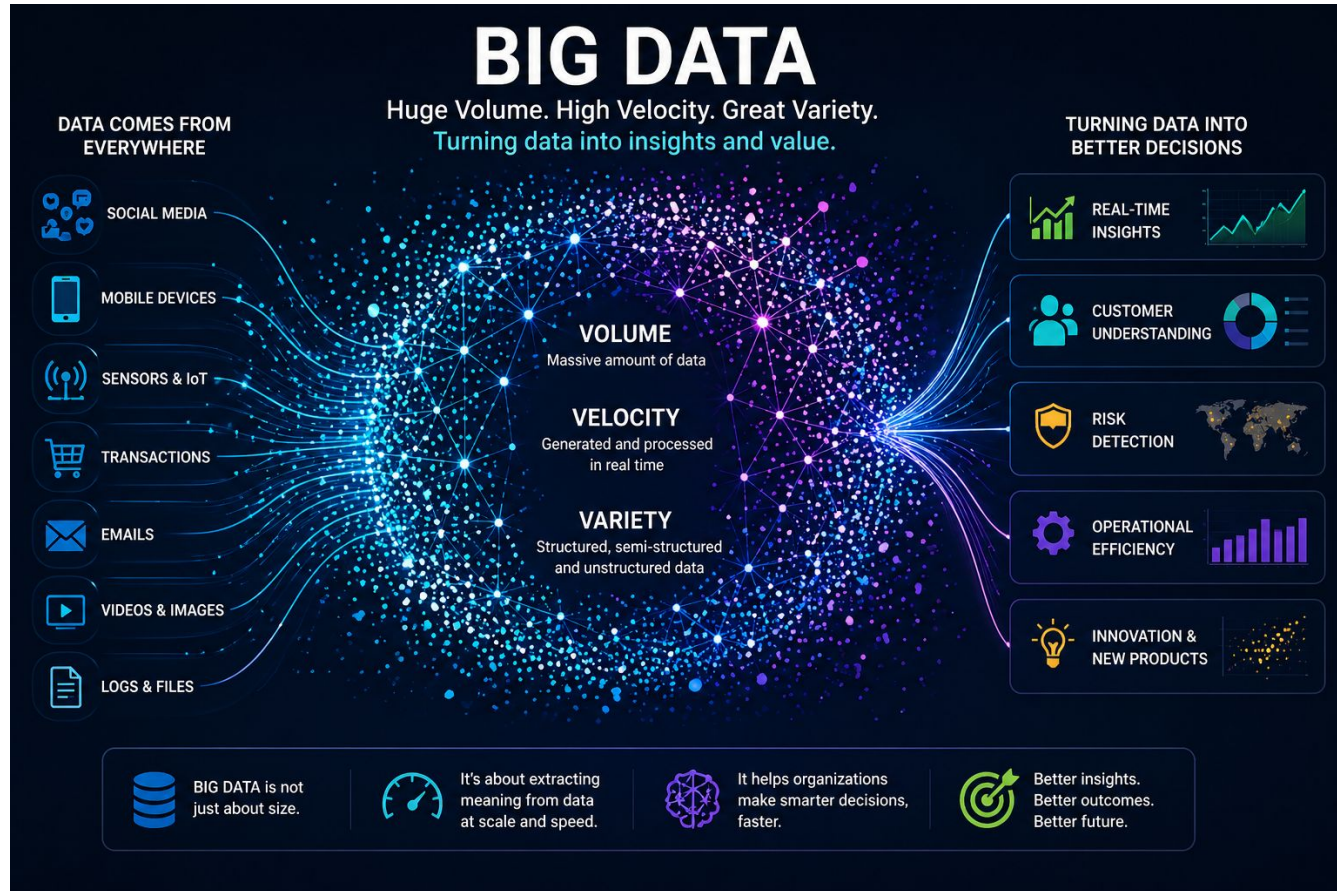
Our Shared Purpose

It isn't just about numbers or code. It's about **asking the right questions** and using data to find answers that matter.

As we dive into this journey together, remember the impact of your findings.

Unlock insights across any field with the power of Data Science.

Why didn't data science exist before?



WHY IS THIS IMPORTANT FOR YOU?



CROSS-INDUSTRY RELEVANCE

Data Science skills are critical across virtually every modern career path.



FUTURE-PROOF SKILL SET

Mastering these competencies makes you highly adaptable and sought-after.



STAY AHEAD OF THE CURVE

Gain a strategic advantage and lead with data-driven insights.

Popular Data Science Languages

Data Science can be done in any language, but a few are more popular.

The Top 3:



Python

Most Popular & Course Focus



Used for everything:
analysis, ML, visualization.



SQL

Data Querying &
Management



R

Statistical Analysis
& Visualization

Also Used:



Julia



Spark



Java/C++



Javascript



For this course, we will focus only on Python.

How Python fits into Data Science

Python is a powerful, easy-to-learn programming language that's used for a wide range of tasks.



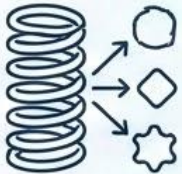
1. Libraries for Everything:

Vast ecosystem of specialized tools for data manipulation, analysis, and visualization (e.g., Pandas, NumPy, Matplotlib).



2. Community Support

Large, active community providing extensive documentation, tutorials, and forums for help.



3. Flexibility

Adaptable for various stages of the data science lifecycle, from data cleaning to deployment.



4. Readability

Clean and intuitive syntax, making code easy to write, understand, and maintain.



Introduction to Github and Github Codespaces

Understanding the Tools



What is Git?
Version Control System

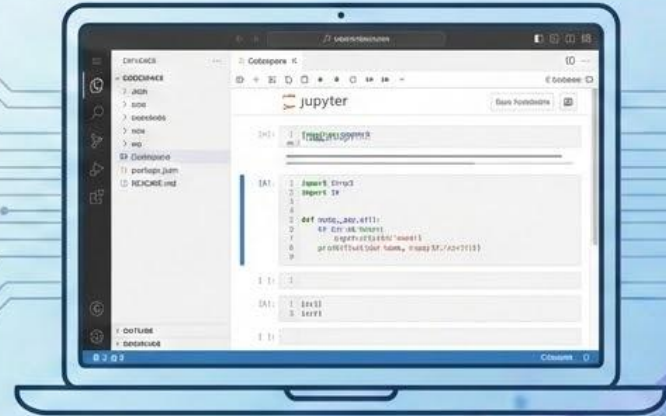


What is Github?
Collaboration Platform



What is Github Codespaces?
Cloud Development Environment

Activity 1: Setting up your GitHub Codespace



1. Follow directions on Schoology & Click [Link](#)
2. Create an account (or log in)
3. Create your first Jupyter Notebook

Activity 2: Explore The Codespaces Environment

Get used to the Codespaces environment.

Ask Questions:

- What do you notice?
- What is something that you aren't sure about?
- How might we go about starting a project from this?



Activity 3: First Jupyter Notebook

Subtitle: Utilizing Markdown & Code Cells for Interactive Data Science

Markdown Cells (Documentation)

Purpose: Writing, Explanations, Planning, Rich Text Formatting.

Features: Headers, bold/italic text, lists, links, images, LaTeX equations.

Example (Rendered):

Data Analysis Plan

- **Step 1:** Load Data
- **Step 2:** Clean Data

Code Cells (Execution)

Purpose: Executing Python code, data analysis, model training.

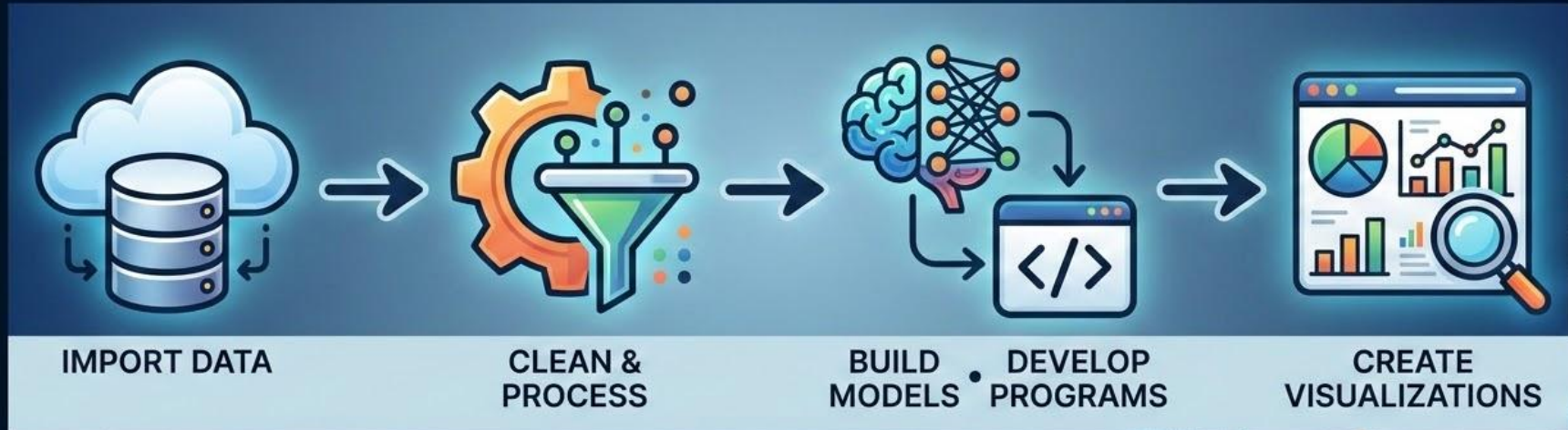
Features: Syntax highlighting, auto-completion, immediate output display.

Example (Output):

[DATA VISUALIZATION PLOT]

Reflection - Why These Tools Matter

These tools are the building blocks we will use to import, clean, process our data and make our models, programs, and visualizations.

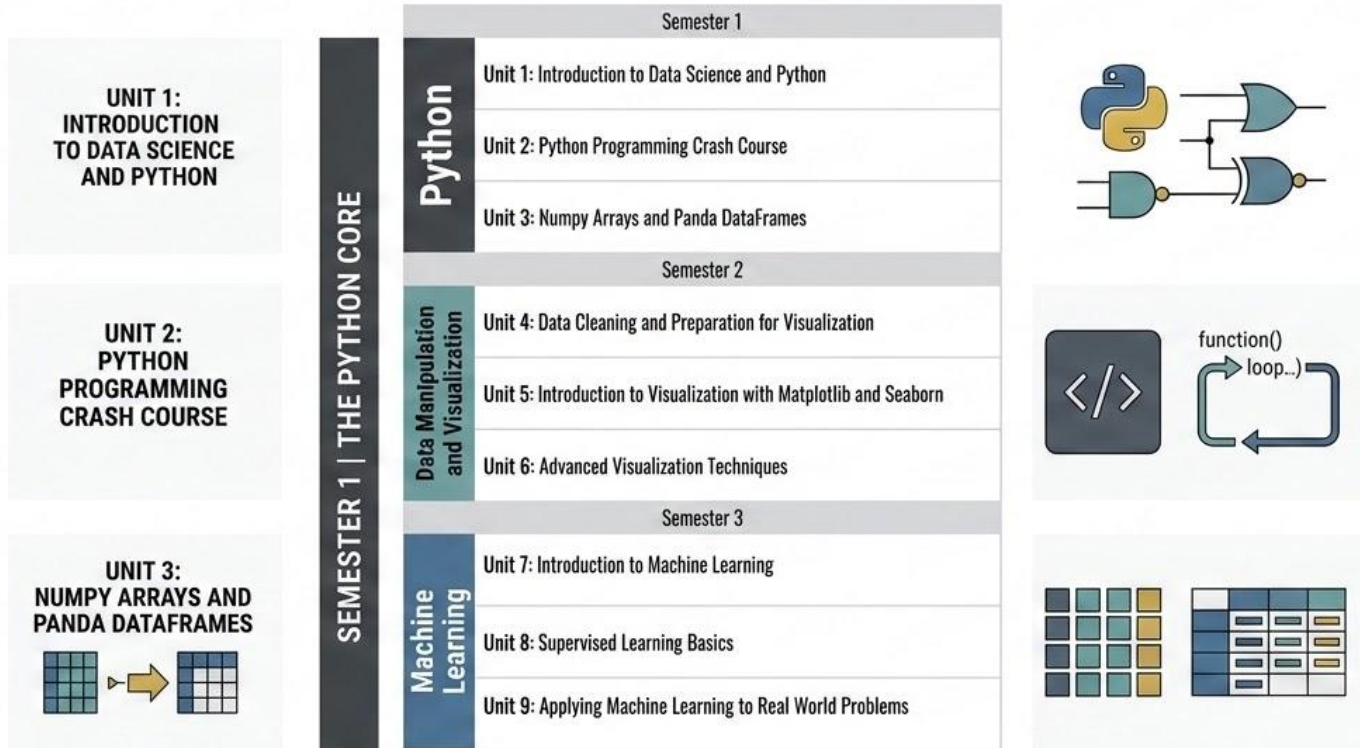


Data Science Workflow: From Raw Data to Actionable Insights



Python Foundation: The Core of Data Science

This is where we are going throughout this course.



Closing: Key Takeaways and Next Steps

Next steps we will start to explore the python language with some simple definitions, programs, and activity.



DEFINITIONS & PYTHON



SIMPLE PROGRAMS



HANDS-ON ACTIVITY